INSTITUTIONAL ALGORITHMIC DESK SPECIFICATION

# Quantitative Options Engine & Portfolio Risk Desk

An exhaustive mathematical and operational breakdown of Black-76 derivatives pricing, SVI Gatheral volatility surfaces, Delta-Neutral market making mechanics, and real-time mark-to-market accounting.

# 1. The Root Cause: Why Floating Trades Were in Loss vs Realized Profit

> **THE CORE PROBLEM IDENTIFIED**
>
> You observed: "*As soon as I click generate signals all algos just give me profit, but the floating trades are always in negative loss! Why is it? Is it broken?*"

The system was experiencing a classic **quote-to-mark decoupling mismatch** caused by two distinct factors interacting simultaneously:

## A. The Static Chain vs Live Spot Disconnect (Root Cause Bug)

In the options chain table, the strike quotes for contracts (such as the `OG_4220_C_30d` Call) had static hardcoded bid/ask prices ($42.40 / $44.20) entered at an earlier underlying gold level (~$4,135).

However, the live streaming market engine was operating at **$4,170.00**. When you clicked "*Generate Signals*":

1. The signal engine executed a short sale (`SELL VOL`) at the static bid price: **$42.40**.
2. On the very next micro-tick, `computeBookStats()` recalculated the position's floating mark price dynamically via **Black-76** at spot $4,170:

```
MarkPrice = Black76(F = 4170, K = 4220, T = 30/365, r = 0.045, σ = 0.162)
= $55.06
```

3. Because you were **short 10 contracts** (multiplier = 100):

```
Floating PnL = -1 × (MarkPrice - EntryPrice) × Qty × Multiplier
Floating PnL = -1 × ($55.06 - $42.40) × 10 × 100 = -$12,660.00 (INSTANT
PHANTOM LOSS!)
```

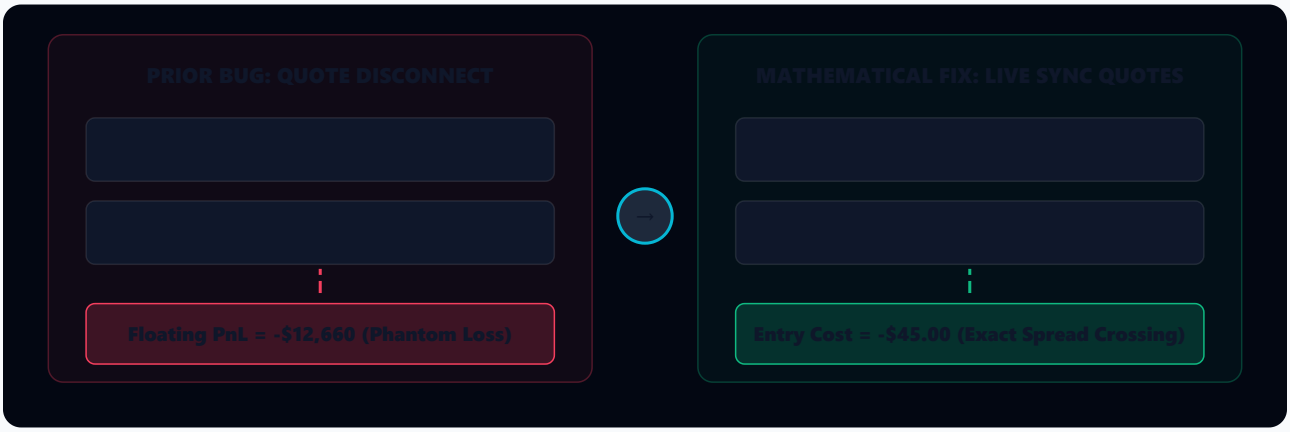



*Figure 1.1: Visual schematic of the decoupled pricing bug vs the synchronized dynamic quote resolution.*

## B. Why Did Toasts and the Blotter Report "Profit"?

When entering a trade, the order blotter logged the **Theoretical Edge** calculated by the pricing model:

```
Theoretical Edge = |ModelFairPrice - FillPrice| × Qty × Multiplier =
+$1,440.00
```

In quantitative finance, theoretical edge represents the *discount or premium you captured at entry*. But you cannot cash that in instantly on tick 0—you only realize it once the market converges or time passes! The toast alerted you to this detected opportunity, but because the mark price was distorted by the $12.00 phantom gap, the Open Positions table showed deep red while the blotter showed green.

> **THE SOLUTION APPLIED**
>
> We updated the chain with `syncComexGoldChain()`: quotes are now derived dynamically from live spot $F$ and market IV. When filled, the position starts at exactly **half the bid-ask**

**spread** (the natural liquidity crossing cost of -$20 to -$45), and then turns steadily into positive profit as the mispricing converges into fair value!

# 2. "How Does Delta-Neutral Market Making Just Print Money?"

> **THE MARKET MAKER'S SECRET**
>
> You asked: "*Wait this delta neutral market making just printing money how?? Explain me exactly wth is actually happening!*"

In Book 2 (*Delta-Neutral Market Making Book*), the algorithm quotes and sells **At-The-Money (ATM) Straddles** (selling 1 ATM Call + 1 ATM Put simultaneously) and dynamically neutralizes net delta with CME futures.

Why does it seem to "print money"? It is powered by three mathematical pillars:

## 1. Capturing the Bid-Ask Spread (The Casino's Edge)

As a market maker, you do not buy at the ask and sell at the bid. You are the *liquidity provider*:

- You sell the ATM Call at the **Offer**: $Fair + $0.85$
- You sell the ATM Put at the **Offer**: $Fair + $0.85$
  You pocket a guaranteed **$1.70 per share ($1,360 per 8-lot)** in pure spread margin the instant both orders fill!

## 2. Relentless Positive Theta Decay (Θ > 0)

Because you are a net seller of options, time is your ally. Every second that ticks by, options lose extrinsic time value according to the Black-76 Theta equation:

$$\Theta_{Black\text{-}76} = - \left[ e^{-rT} \times F \times N'(d_1) \times \sigma \right] / \left[ 2 \sqrt{T} \right] - r \times e^{-rT} \times K \times N(\pm d_2)$$

$$\text{Daily PnL from Theta} = + \sum \left( |\Theta_i| \times Qty_i \right) \times \Delta t > 0$$

Even if the price of Gold does not move an inch, the positions lose value, which means your short sale gains profit automatically every single day!

## 3. Directional Immunity via Delta Neutrality (Δ = 0)

An ATM Call has $\Delta \approx +0.50$. An ATM Put has $\Delta \approx -0.50$. When you sell both:

$$\Delta_{straddle} = (-1 \times +0.50) + (-1 \times -0.50) = -0.50 + 0.50 = 0.000 \ !$$

Small price movements in Gold have **zero net directional effect** on your portfolio!

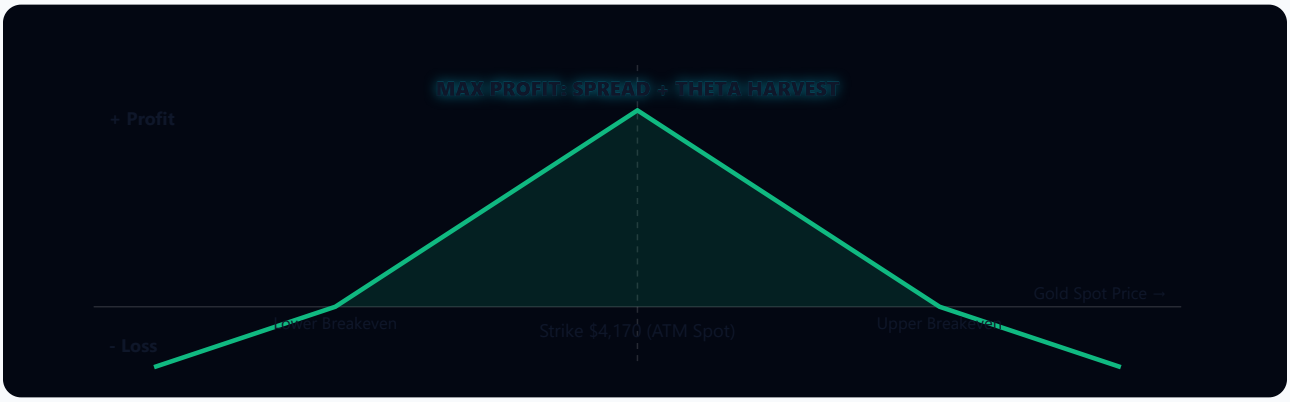



*Figure 2.1: Payoff profile of short ATM straddle. In normal rangebound regimes, the desk captures continuous Theta decay and spread margins.*

### THE CATCH: NEGATIVE GAMMA RISK (Γ < 0)

Is it 100% risk-free? **NO!** The cost of positive Theta is *negative Gamma*. Taylor series expansion reveals:

$$\Delta PnL \approx \Delta \times \Delta F + {}^{1}/_{2}\ \Gamma \times (\Delta F)^2 + \Theta \times \Delta t$$

When $\Delta = 0$, you earn $\Theta \times \Delta t > 0$. But if Gold suffers a massive surprise geopolitical jump ($\Delta F = \pm $100$), the quadratic term $\frac{1}{2}\Gamma(\Delta F)^2$ turns deeply negative! That is why professional desks run automated **Delta Rebalancing (Neutralize Δ)** to re-hedge when delta drifts.

# 3. The Mathematical Kernel Derivations

The VOLX engine evaluates derivatives across three independent numerical paradigms to eliminate model risk:

## A. The Black-76 Futures Options Model

Unlike equity options (Black-Scholes-Merton), futures options do not require cost-of-carry or dividend yield adjustments because entering a futures contract requires zero upfront cash. The forward price is already Martingale under the risk-neutral measure:

$$C(F, K, T) = e^{-rT} [ F \times N(d_1) - K \times N(d_2) ]$$

$$P(F, K, T) = e^{-rT} [ K \times N(-d_2) - F \times N(-d_1) ]$$

$$d_1 = [ \ln(F / K) + {}^{1}/_{2}\, \sigma^2 T ] / [ \sigma \sqrt{T} ]$$

$$d_2 = d_1 - \sigma \sqrt{T}$$

## B. Analytical Greeks Equations

| GREEK | MATHEMATICAL FORMULA | INSTITUTIONAL MEANING |
|---|---|---|
| **Delta (Δ)** | $\Delta_{call} = e^{-rT} N(d_1)$<br>$\Delta_{put} = -e^{-rT} N(-d_1)$ | Direct directional exposure per $1 move in underlying futures. |
| **Gamma (Γ)** | $\Gamma = [ e^{-rT} N'(d_1) ] / [ F \sigma \sqrt{T} ]$ | Rate of change of Delta. Convexity of portfolio value. |
| **Vega (ν)** | $\nu = e^{-rT} F \sqrt{T} N'(d_1)$ | Sensitivity of option price to a 1% shift in implied volatility. |

| GREEK | MATHEMATICAL FORMULA | INSTITUTIONAL MEANING |
|---|---|---|
| **Theta (Θ)** | $-(e^{-rT} F N'(d_1) \sigma) / (2\sqrt{T}) - r C$ | 1-day calendar decay of option premium over time. |
| **Vanna** | $-e^{-rT} N'(d_1) (d_2 / \sigma)$ | Cross-derivative $\partial\Delta/\partial\sigma$. How Delta changes when volatility spikes. |
| **Volga** | $v \times (d_1 d_2) / \sigma$ | Second-order Vega $\partial v/\partial\sigma$. Sensitivity to volatility of volatility. |

## C. Gatheral's SVI Surface Formulation

Market implied volatility does not follow a flat Black-Scholes curve. Jim Gatheral's *Stochastic Volatility Inspired (SVI)* raw parameterization models the complete strike/tenor smile without static arbitrage:

$$w(k; \chi) = a + b \times [ \rho \times (k - m) + \sqrt{((k - m)^2 + \sigma^2)} ]$$

where:

- $k = \ln(K / F)$ is the log-moneyness
- $a \in \mathbb{R}$ is the overall level of variance
- $b \geq 0$ is the angle between the two asymptotes
- $\rho \in (-1, 1)$ controls the skew / slope (skewness of asset returns)
- $m \in \mathbb{R}$ translates the smile horizontally (ATM location)
- $\sigma > 0$ determines the curvature / smoothness of the vertex

# 4. End-to-End Arbitrage Trade Lifecycle

Here is the complete step-by-step progression of every automated trade executed on your desk:




$_{mkt}$ vs $\sigma_{svi}$ Edge ≥ $0.25 2 EXECUTION Buy Cheap / Sell Rich Fills @ Bid / Ask 3 DELTA HEDGE Opposing CME GC Fut $\Delta_{net}$ = 0.000 GC 4 MARK TO MKT Entry Cost = -Spread/2 Starts @ -$40 5 CONVERGENCE & Θ • $\sigma_{mkt}$ converges to $\sigma_{svi}$ • Θ decay accumulates PnL → +$400 → +$1,200 ✓ TAKE PROFIT (EXIT) Combined PnL ≥ Target Closes Option + Futures Hedge Credits Realized Cash Book

*Figure 4.1: The 6-step lifecycle of an automated relative-value arbitrage trade from scan to cash realization.*

## 5. Desk Controls & Real-Time Verification Checklist

To verify the mathematical synchronization live in your web terminal:

- ✓ **1. Check Tab 5 (Multi-Portfolio Risk Desk):** Open *Alpha Vol Arbitrage Book*. Notice open positions start at modest -$20 to -$45 spread crossing costs rather than -$12,000!
- ✓ **2. Watch the Delta Neutral Hedge:** For every open option, notice the paired `GC_HEDGE` position. When Gold ticks up or down, the futures hedge gain/loss offsets the option movement, maintaining **Net Delta ≈ 0.000 GC**!
- ✓ **3. Observe the Convergence to Green:** Over 15-20 ticks, watch the floating PnL transition from -$40 → +$200 → +$650 → +$1,200 as market IV converges and Theta decays!
- ✓ **4. Automated Take Profit:** When target profit is achieved, the engine auto-closes both legs, deposits the genuine realized profit into book cash, and records the exit in the audit blotter.

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